DATA ITEM DESCRIPTION

Form Approved
OMB No 0704 0188

1. TITLE	2 IDENTIFICATION NUMBER
REPAIR WELDING APPROVAL RECORD FOR FORGINGS OR CASTINGS	DI-FORG-81195

3. DESCRIPTION/PURPOSE

3.1 The Repair Welding Approval Record for Forgings or Castings provides repair welding process and welder qualifications data.

3.2 The Repair Welding Approval Record for Forgings or Castings data is used to determine whether repair welding of forgings or castings should be authorized.

4 APPROVAL DATE (YYMMDD)	5 OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a DTIC APPLICABLE	6b GIDEP APPLICABL
910502	MR		

7. APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) satisfies the requirements of paragraph 3.5.2 of MIL-S-46172A.

7.2 This Data Item Description supersedes DI-P-1637

8. APPROVAL LIMITATION	9a APPLICABLE FORMS	9b. AMSC NUMBER
		A6119

10 PREPARATION INSTRUCTIONS

10.1 Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

10.2 Format. The Repair Welding Approval Record For Forgings or Castings format shall be contractor selected. Unless effective presentation would be degraded the initially used format arrangement shall be used for all subsequent submissions.

10.3 Content. The Repair Welding Approval Record For Forgings or Castings content shall be in accordance with the elements of Table I and the Supplementary Data as follows:

10.3.1 RECORD NO. Contractor assigned (sequential and non-duplicative) record number with appropriate revision letter.

10.3.2 Forging/Casting Name. Name of the forging or casting.

10.3.3 Dwg. (Drawing) No. Applicable drawing number.

10.3.4 Contract No. Applicable contract number.

10.3.5 Date. Record preparation data.

11 DISTRIBUTION STATEMENT

(Continued on Page 2)

DISTRIBUTION STATEMENT A: Approved for public release: distribution is unlimited.

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10. PREPARATION INSTRUCTIONS (Continued)

10.3.6 WELDING PROCESS. Identify the welding process.

10.3.7 BASE METALS. Identify the Material Specification, the Type or Grade of the base metal(s), the Thickness Range, and the Diameter.

10.3.8 FILLER MATERIAL. When applicable: Identify the Weld Metal Type, the Size of the Electrode or Filler, the MIL (military) Specification, the AWS (American Welding Society) Classification, the Electrode Flux Class, and the Consumable Insert.

10.3.9 GAS. When applicable: Identify the Type of Gas or Gases, Composition of Gas Mixture, Orifice or Gas Cup Size, Shielding Gas, Flow Rate, Gas Backing, and Trailing Shielding Gas Composition.

10.3.10 POSITION. Identify the Position of Groove, Weld Progression (Uphill, Downhill), Method of Backgouging.

10.3.11 PREHEAT. Identify the Preheat Temp (temperature), Interpass Temp (temperature), and Preheat Maintenance.

10.3.12 Electrical Characteristics. Identify the Current (AC or DC), Polarity (Straight or Reversed), Amps (amperage), and Volts (voltage).

10.3.13 TECHNIQUE. When applicable: Identify the technique Travel Speed, String or Weave Bead, Oscillation, Multipass or Single Pass (how many passes per side), Single or Multiple Electrodes, Initial and Interpass Cleaning.

10.3.14 Post Heat. When applicable: Identify the post heat Temperature and Time.

10.3.15 Heat Treatment After Welding. Identify the heat treatment after welding temperature (degrees Fahrenheit), time (minutes) and Quench Media for both Austenitizing, and tempering.

10.3.16 Manufacturer. The name of the manufacturer (contractor).

10.3.17 Signature. Appropriate manufacturer's (contractor's) representative's signature.

10.4 Supplementary Data. The following supplementary data shall be included:

10.4.1 Weld size and location. A dimensioned sketch illustrating the size and location of the weld(s).

10.4.2 Weld repair procedure. Complete information on the weld repair procedure.

10.4.3 Welder qualifications. Complete information on the welder's qualifications.

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TABLE I. Weld Repair Procedure & Welder Qualification.

RECORD NO. _____

Forging/Casting Name_____
Dwg. No._____
Contract No._____
Date_____

WELDING PROCESS_____

TYPES_____
(Manual, Automatic, Semi-Auto)

BASE METALS

Material Spec._____
Type or Grade_____
Thickness Range_____
Diameter_____

FILLER MATERIAL

Weld Metal Type_____
Size of Electrode_____
or Filler_____
MIL Specification_____
AWS Classification_____
Electrode Flux Class_____
Consumable Insert_____
Other_____

GAS

Type of Gas or Gases_____
Composition of Gas Mixture_____
Orifice or Gas Cup Size_____
Shielding Gas_____
Flow Rate_____
Gas Backing_____
Trailing Shielding Gas
Composition_____
Other_____

POSITION

Position of Groove_____
Weld Progression_____
(Uphill, Downhill)
Other_____
Methods of Backgouging_____
Other_____

PREHEAT

Preheat Temp_____
Interpass Temp_____
Preheat Maintenance_____
Other_____

ELECTRICAL CHARACTERISTICS

Current (AC OR DC)_____
Polarity (Straight or Reverse)____
Amps_____Volts_____
Other_____

TECHNIQUE

Travel Speed_____
String or Weave Bead_____
Oscillation_____
Multipass or Single
Pass_____
(per side)
Single or Multiple
Electrodes_____
Initial & Interpass Cleaning
(Brushing, Grinding, etc.)

POST HEAT

Temperature_____
Time_____
Other_____

HEAT TREATMENT AFTER WELDING

Austenitize_____^{o}F.____minutes
Quench Media_____
Tempering_____^{o}F.____minutes
Quench Media_____

Manufacturer_____

Signature_____

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